

Question	Answer	Marks
4(a)	ΔU : <u>increase</u> in internal energy (of the system)	B1
	Q: thermal energy transferred <u>to</u> the system	B1
4(b)(i)	$W = p\Delta V$	C1
	$= (2.0 \times 10^5) \times (0.048 - 0.012) = 7200 \text{ J}$	A1
4(b)(ii)	<i>either:</i> volume decreases so W is positive <i>or:</i> gas is compressed so W is positive	B1
4(b)(iii)	<i>either:</i> volume is proportional to (thermodynamic) temperature and (thermodynamic) temperature is proportional to internal energy	B1
	volume quarters so internal energy quarters	B1
	<i>or:</i> volume quarters so (thermodynamic) temperature quarters	(B1)
	(thermodynamic) temperature quarters so internal energy quarters	(B1)

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Question	Answer	Marks
4(c)	$\Delta U = 3.6 - 14.4$ $= -10.8 \text{ kJ}$	C1
	$Q = -10.8 - 7.2$ $= \mathbf{-18.0 \text{ kJ}}$	A1